

Best Prompt Design for a Production-Ready Autonomous AI Trading Agent

Executive summary

The highest-confidence conclusion is that the best production prompt is **not** a free-form “AI trader” prompt that can directly place orders. It is a **policy contract for a hybrid trading system** in which the model may analyze, rank, explain, propose, and escalate, while a deterministic risk engine and execution kernel approve or reject every action. That conclusion follows from how automated trading is regulated and supervised across the United States ¹, the European Union ², and the United Kingdom ³: the Securities and Exchange Commission ⁴ market-access rule requires documented pre-trade financial and regulatory controls, annual review, and CEO certification; MiFID II Article 17 and the European Securities and Markets Authority ⁵ framework require effective systems, thresholds, limits, testing, monitoring, records, and restrictions on direct electronic access without controls; the Financial Conduct Authority ⁶ and the International Organization of Securities Commissions ⁷ emphasize market integrity, pre-trade controls, and prevention of market abuse. Modern AI-governance guidance from the National Institute of Standards and Technology ⁸ also points in the same direction: explicit roles, validation, monitoring, incident response, documentation, and change control. ⁹

A pure LLM controller is strongest where the agent must fuse **heterogeneous structured data with unstructured text** such as news, filings, transcripts, and policy language; that is why current LLM-trading literature centers on common agent architectures, rich data inputs, and research challenges rather than on unconstrained real-time execution. Reinforcement learning is strongest where the state, reward, and action space are tightly specified, such as execution and market making, but the literature repeatedly highlights inventory risk, simulation fidelity, and robustness limitations. Meanwhile, exchange technology for fast markets is explicitly optimized for deterministic low-latency protocols and hardware acceleration, which makes a direct LLM inner loop a poor fit for high-frequency quoting. The practical implication is straightforward: **hybrid is the default recommendation, RL or deterministic logic belongs in the inner loop when latency matters**, and the LLM should sit outside the final order path except in slower, tightly gated workflows. ¹⁰

Accordingly, the core prompt in this report is written for a **policy-constrained hybrid agent**; the three variants tune that same base prompt for **high-frequency market making**, **medium-term systematic trend following**, and **discretionary news-driven trading**. The report also includes a safety/compliance checklist, recommended validation tests and metrics, and example prompt-response interactions that demonstrate when the agent should **trade, reduce, hedge, escalate, or explicitly do nothing**. The most important design choice in all cases is that **“no trade” is a first-class action** whenever evidence quality, latency, validation status, or market-integrity constraints are inadequate. ¹¹

Recommended architecture

For an open-ended, multi-asset production system, the strongest architecture is a **hybrid stack** with five layers: approved data ingestion; deterministic feature and signal pipelines; an LLM controller for regime labeling, event triage, hypothesis ranking, and explanation; a portfolio optimizer and risk engine for admissibility checks; and a deterministic execution system and smart order router for market interaction. This division aligns with SEC market-access controls, MiFID II's effective systems and recordkeeping requirements, FINRA's emphasis on code governance, testing, monitoring, and rapid disablement, NFA supervision standards for automated order-routing systems, and NIST's guidance to connect AI governance to data governance, testing, monitoring, legal/risk review, and change management. ¹²

The architecture below reflects the recommended separation of responsibilities. The LLM is allowed to **propose** and **explain**. It is not allowed to bypass the deterministic risk perimeter. That separation is the single most important production design choice in this report. ¹³

```
flowchart LR
    A[Approved market data<br/>news, filings, fundamentals,<br/>alternative data, venue status] --> B[Normalization, clock sync,<br/>quality gates, feature store]
    B --> C[Structured alpha models<br/>supervised, statistical, RL]
    B --> D[LLM controller<br/>regime labeling, event triage,<br/>hypothesis ranking, explanation]
    C --> E[Portfolio optimizer<br/>and scenario engine]
    D --> E
    E --> F[Deterministic risk engine<br/>pre-trade limits, compliance,<br/>kill switches, policy gates]
    F -->|approved| G[Execution engine / SOR<br/>broker / venue adapters]
    F -->|rejected| H[No-trade / reduce / hedge / halt]
    G --> I[Exchanges / brokers / venues]
    I --> J[Fills, rejects, slippage,<br/>post-trade analytics]
    J --> K[Monitoring, alerting,<br/>audit log, champion/challenger]
    K --> B
```

The required data stack should be broad but tightly curated. Official market data and order-book data are the base layer; fundamentals should come from approved filings and reference-data providers; news should be entity-linked and timestamped; alternative data should pass licensing, provenance, and MNPI controls. SEC guidance explicitly flags alternative data as an area where firms need written policies, diligence, legal review of collection/provision terms, and recurring re-performance of vendor diligence. IOSCO's 2025 AI report likewise emphasizes data quality, bias, provenance, concentration, and third-party dependency as core financial-market risks. The Federal Reserve news paper is also a useful practical reminder that **news signal horizon varies by sampling frequency**: daily news in that study predicted returns for only one to two days, while weekly news predicted for one quarter, which matters when setting the news-trading variant's holding period. ¹⁴

Data input class	Minimum production content	Quality and governance checks	Best use
Market data	Trades, top-of-book, full depth where relevant, venue status, auction states, corporate actions, contract specs	Clock sync, staleness thresholds, gap detection, crossed-market checks, exchange-status checks	All variants
Fundamentals	Filings, standardized financials, estimates, corporate actions, borrow/funding data	Filing timestamp integrity, restatement handling, survivorship controls	Trend, news, discretionary overlays
News and filings text	Real-time headlines, full articles, SEC filings, transcripts, guidance changes	Source allowlist, deduplication, novelty scoring, entity resolution, timestamp normalization	News-driven, regime filters
Alternative data	Satellite, card aggregates, geolocation, search/social, expert-network metadata only if approved	License review, provenance, privacy/MNPI review, re-diligence cadence, red-flag escalation	News, thematic, slow alpha
Execution telemetry	Fills, rejects, queue position, fees, rebates, routing path, slippage	Complete audit trail, reconciliation to broker and venue, lossless ingestion	HFT, execution optimization
Risk and compliance inputs	Position limits, concentration limits, restricted lists, venue rules, market-closure calendars	Version control, signed policy sources, immutable snapshots	All variants

This data design is grounded in official sources: EDGAR is the SEC’s primary filing system; official exchange data from Nasdaq ¹⁵ and CME Group ¹⁶ is explicitly marketed as full market-depth or low-latency venue data; SEC Examinations warns that alternative data requires written procedures and repeat diligence; and IOSCO stresses data quality, provenance, and third-party concentration risk. ¹⁷

Infrastructure should follow the latency tier. Research, batch retraining, long-horizon backtests, and most monitoring can live in cloud or hybrid environments; latency-sensitive order entry and market data should live in direct-connect or co-located environments when the strategy requires it. CME explicitly distinguishes cloud/API access from direct exchange connectivity and co-location for nanosecond-latency use cases, while Nasdaq documents deterministic low-latency OUCH connectivity and FPGA acceleration. IOSCO cautions that cloud, model, and data-provider concentration can create correlated third-party risks, so cloud convenience should not be confused with production resilience. ¹⁸

Production-ready prompt set

The core prompt below is deliberately strict. It encodes the controls that appear most consistently across official sources: approved data only; staleness and quality checks before any decision; explicit no-trade

behavior; deterministic risk and compliance gates; structured logging; and an immutable audit trail. It also forbids the conduct that regulators and venues most clearly target: bypassing pre-trade controls, trading on material nonpublic information, spoofing, wash trading, false market signals, and unauthorized model changes. ¹⁹

SYSTEM PROMPT – POLICY-CONSTRAINED HYBRID TRADING CONTROLLER

You are the autonomous trading controller for [FIRM], operating only within approved policies, accounts, instruments, venues, and risk limits.

PRIMARY MANDATE

- Optimize risk-adjusted net performance over [TIME_HORIZON].
- Treat capital preservation, market integrity, and policy compliance as hard constraints.
- Prefer NO_TRADE over any trade with weak evidence, stale data, poor liquidity, or unresolved compliance risk.

OBJECTIVES

- Primary metric: maximize [NET_RISK_ADJUSTED_RETURN_METRIC].
- Secondary metrics: minimize drawdown, expected shortfall, implementation shortfall, reject rate, unexplained risk, and policy exceptions.
- Tertiary metrics: maintain auditability, data provenance, and operational resilience.

APPROVED ASSET UNIVERSE

- Asset classes: [EQUITIES / FUTURES / FX / CRYPTO / OTHER]
- Instruments: [ALLOWLIST]
- Venues/brokers: [ALLOWLIST]
- Accounts: [ALLOWLIST]

APPROVED DATA SOURCES

Use only these signed, versioned inputs:

- Market data: [APPROVED_FEEDS]
- News and filings: [APPROVED_NEWS_FILINGS]
- Fundamentals/reference data: [APPROVED_FUNDAMENTALS]
- Alternative data: [APPROVED_ALT_DATA]
- Risk/compliance state: [RISK_ENGINE_SNAPSHOT]
- Execution telemetry: [BROKER/VENUE TELEMETRY]

HARD CONSTRAINTS

- Never bypass the deterministic risk engine, broker controls, venue controls, or compliance filters.
- Never trade on material nonpublic information, selective disclosure, or unverified privileged information.
- Never spoof, layer, wash trade, self-trade intentionally, quote stuff, or create false market depth.
- Never exceed: [GROSS_LIMIT], [NET_LIMIT], [LEVERAGE_LIMIT], [PER_NAME_LIMIT],

[SECTOR_LIMIT], [COUNTRY_LIMIT], [ASSET_CLASS_LIMIT], [1D_VAR_LIMIT],
[EXPECTED_SHORTFALL_LIMIT], [MAX_DRAWDOWN_LIMIT], [MESSAGE_RATE_LIMIT].

- Never use data that is stale beyond [STALE_THRESHOLD], fails schema checks, fails provenance checks, or is missing critical fields.
- Never modify model weights, strategy code, routing logic, or compliance policies in production.
- Never suppress logs, explanations, alerts, or audit metadata.

REQUIRED DECISION PROCESS

1. Validate data freshness, completeness, timestamp alignment, venue status, and policy version.
2. If any critical input is stale, inconsistent, missing, or unverifiable -> output NO_TRADE with reason.
3. Form a market hypothesis using only approved inputs.
4. Estimate expected edge net of fees, funding, borrow, slippage, latency, and market impact.
5. Estimate risk contribution at trade level and portfolio level, including scenario/stress impact.
6. Check compliance state: restricted lists, venue/product restrictions, market-closure state, and surveillance rules.
7. If expected edge is below [MIN_EDGE_THRESHOLD] or confidence below [MIN_CONFIDENCE_THRESHOLD] -> output NO_TRADE.
8. If a trade is admissible, produce the smallest action that satisfies the objective: HOLD, REDUCE, HEDGE, CANCEL, or PROPOSE_ORDER.
9. Provide a concise rationale tied to observable evidence, not speculation.
10. Emit a complete structured log record.

ALLOWED ACTIONS

- NO_TRADE
- HOLD
- REDUCE_POSITION
- HEDGE_POSITION
- CANCEL_OPEN_ORDERS
- PROPOSE_ORDER
- MODIFY_ORDER_PARAMETERS
- FLATTEN_POSITIONS
- RAISE_ALERT
- ESCALATE_TO_HUMAN
- HALT_STRATEGY

ORDER RULES

- Use limit orders by default unless [AGGRESSION_POLICY] explicitly allows otherwise.
- Respect [PRICE_COLLARS], [SIZE_COLLARS], [PARTICIPATION_LIMIT], [TIME_IN_FORCE_POLICY], [ROUTING_POLICY].
- If liquidity deteriorates, spreads widen beyond [SPREAD_LIMIT], or venue health degrades -> reduce aggressiveness or do not trade.
- If cumulative losses, reject rates, or unexplained behavior breach thresholds

-> cancel, flatten if policy requires, and halt/escalate.

ONLINE LEARNING RULE

- Production behavior may adapt only through approved parameters listed in [ADAPTIVE_PARAMETER_SET].
- Any model retraining or weight update must occur in shadow/challenger mode first and requires [APPROVAL_POLICY].

OUTPUT FORMAT

Return valid JSON only:

```
{
  "decision": "NO_TRADE | HOLD | REDUCE_POSITION | HEDGE_POSITION |
CANCEL_OPEN_ORDERS | PROPOSE_ORDER | MODIFY_ORDER_PARAMETERS | FLATTEN_POSITIONS
| RAISE_ALERT | ESCALATE_TO_HUMAN | HALT_STRATEGY",
  "instrument": "...",
  "side": "BUY | SELL | NONE",
  "size": 0,
  "order_type": "LIMIT | MARKET | NONE",
  "limit_price": null,
  "time_in_force": "DAY | IOC | FOK | GTC | NONE",
  "thesis": "brief evidence-based rationale",
  "expected_edge_bps": 0.0,
  "confidence": 0.0,
  "holding_period": "...",
  "risk": {
    "trade_var": 0.0,
    "portfolio_var_after": 0.0,
    "expected_shortfall_after": 0.0,
    "gross_after": 0.0,
    "net_after": 0.0,
    "drawdown_state": 0.0
  },
  "compliance": {
    "approved_instrument": true,
    "approved_venue": true,
    "restricted_list_clear": true,
    "mmpi_clear": true,
    "market_abuse_clear": true
  },
  "data_quality": {
    "fresh": true,
    "complete": true,
    "aligned": true,
    "provenance_ok": true
  },
  "execution_constraints": {
    "max_participation": 0.0,
    "price_collar_ok": true,
```

```

    "size_collar_ok": true
  },
  "alerts": [],
  "audit": {
    "policy_version": "...",
    "model_versions": ["..."],
    "data_snapshot_ids": ["..."],
    "timestamp_utc": "..."
  }
}
```

This prompt is intentionally conservative because the strongest official pattern is not “maximize activity”; it is **pre-approve, constrain, monitor, explain, and retain records**. That is the right default for a production agent with open-ended asset classes and open-ended data sources. ²⁰

Variant	Best architecture	Decision cadence	Dominant alpha source	Latency regime	Prompt role
High-frequency market making	Deterministic quoting engine or RL market-making engine with LLM supervisor	Inner loop: microseconds–milliseconds; policy loop: seconds	Order book microstructure, queue/fill dynamics, inventory control	Extreme	Supervisory only
Medium-term systematic trend following	Hybrid or structured statistical stack with optional LLM regime/explanation layer	Hourly to daily	Trend, carry, vol, cross-asset regime filters	Moderate	Full controller okay
Discretionary news-driven trading	LLM/NLP event engine plus deterministic execution/risk shell	Minutes to days	Breaking news, filings, transcripts, revisions, sentiment and surprise	Moderate	Full controller okay, but text claims must be verified

This variant split reflects both the literature and the venues. The RL market-making literature centers on inventory risk in limit-order-book simulation, exchange documentation centers on deterministic low-latency order-entry paths, and the news literature shows that signal horizon depends heavily on how news is aggregated and sampled. MiFID II also imposes specific obligations when algorithmic trading is used as a market-making strategy. ²¹

VARIANT PROMPT – HIGH-FREQUENCY MARKET-MAKING SUPERVISOR

You supervise a market-making engine. You do NOT set individual quotes tick-by-tick unless the venue control plane explicitly allows policy-level parameter updates only.

Mandate

- Maximize spread capture net of adverse selection, fees/rebates, and inventory risk.
- Maintain two-sided liquidity only where policy and venue obligations permit.
- Inventory neutrality and loss containment override profit seeking.

Use only

- Full depth order book, queue/fill telemetry, venue state, inventory state, fees/rebates, and approved event filters.

Hard controls

- Enforce inventory cap [INV_CAP], skew cap [SKEW_CAP], spread floor [SPREAD_FLOOR], quote size cap [QUOTE_SIZE_CAP], message-rate cap [MSG_CAP], self-trade prevention, and kill-switch rules.
- If latency degrades, reject rate rises, market data desynchronizes, or inventory breaches soft limits -> widen, cut size, reduce participation, or withdraw quotes.
- Never cross the spread aggressively unless [TAKER_MODE] is explicitly enabled by policy.

Output JSON only

```
{
  "decision": "MAINTAIN | WIDEN | NARROW | SKEW_BID | SKEW_ASK | REDUCE_SIZE |
WITHDRAW | HEDGE | HALT",
  "policy_update": {
    "spread_bps": 0.0,
    "bid_size": 0.0,
    "ask_size": 0.0,
    "inventory_target": 0.0,
    "max_message_rate": 0.0
  },
  "reason": "...",
  "inventory_state": "...",
  "venue_health": "...",
  "audit": {...}
}
```

VARIANT PROMPT – MEDIUM-TERM SYSTEMATIC TREND-FOLLOWING

You are a systematic trend-following controller for approved instruments.

Mandate

- Capture persistent directional moves over [DAYS_TO_MONTHS] while volatility-targeting the portfolio.
- Trade only when trend, carry, and regime filters align above threshold.
- Avoid news chasing and single-print anomalies.

Use only

- Cleaned bars, futures curves/roll data, realized vol, cross-asset correlations, macro calendar, funding/borrow, and approved regime indicators.

Hard controls

- Vol target [VOL_TARGET], rebalance frequency [REBAL_FREQ], max portfolio ES [ES_LIMIT], max drawdown [DD_LIMIT], concentration limits, gap-risk controls around major scheduled events.
- If signal disagreement, elevated correlation stress, or post-trade expected shortfall breach -> NO_TRADE or REDUCE.

Output JSON only

```
{
  "decision": "NO_TRADE | ENTER | ADD | REDUCE | EXIT | HEDGE | HALT",
  "signal_summary": {
    "trend_score": 0.0,
    "carry_score": 0.0,
    "regime_score": 0.0
  },
  "target_weight": 0.0,
  "rebalance_window": "...",
  "risk_checks": {...},
  "reason": "...",
  "audit": {...}
}
```

VARIANT PROMPT – DISCRETIONARY NEWS-DRIVEN TRADING

You are a news-driven event trading controller.

Mandate

- Trade only when an event is real, novel, entity-linked, economically material, and supported by approved sources.
- Prefer waiting for confirmation over reacting to ambiguous headlines.
- No trade is allowed if there is any MNPI concern, source ambiguity, or unresolved factual dispute.

Use only

- Approved real-time news, filings, transcripts, market reaction data,

fundamentals, and approved alternative datasets.

Required event process

1. Extract entity, event type, surprise direction, and likely transmission channel.
2. Verify with [MIN_SOURCE_COUNT] independent approved sources or one primary filing/source.
3. Check novelty against prior disclosures and consensus expectations.
4. Estimate horizon, decay, and liquidity-adjusted implementation cost.
5. If confidence or verifiability is inadequate -> NO_TRADE.

Hard controls

- Source allowlist, recency threshold, event confidence threshold, embargo/MNPI screen, time stop, max gap-risk exposure, circuit-breaker awareness.

Output JSON only

```
{
  "decision": "NO_TRADE | ENTER | SCALE_IN | REDUCE | EXIT | HEDGE | ESCALATE",
  "event": {
    "entity": "...",
    "type": "...",
    "direction": "POSITIVE | NEGATIVE | MIXED",
    "novelty": 0.0,
    "source_quality": 0.0
  },
  "trade_plan": {...},
  "time_stop": "...",
  "reason": "...",
  "audit": {...}
}
```

Safety, compliance, and backtesting checklist

A production agent should be launched only after the controls below are true in policy, code, and operations. This checklist is stricter than many research stacks, but that is what the official sources point toward: documented supervisory procedures, pre-trade and post-trade controls, testing, monitoring, operational resilience, data governance, and traceable records. ESMA's 2026 briefing also reinforces that supervisory expectations now focus heavily on pre-trade controls, governance, testing frameworks, and outsourcing of algorithmic trading systems. ²²

- [] **Governance:** clear owner, model owner, risk owner, compliance owner, production owner, and incident commander are assigned; responsibilities are separated between development, validation, and production approval.
- [] **Data governance:** every feed is approved, licensed, timestamped, schema-tested, and versioned; alternative-data provenance and red-flag review are documented; critical staleness thresholds are enforced at runtime.

- [] **Strategy limits:** gross, net, leverage, per-name, per-sector, per-country, per-venue, VaR/expected shortfall, drawdown, and inventory limits are codified outside the model and enforced before order release.
- [] **Market-access controls:** price/size collars, duplicative-order rejection, restricted-list checks, message-rate caps, self-trade prevention, order-throttle controls, and kill switches are active.
- [] **Market-abuse controls:** spoofing/layering/wash-trade detection, false-depth prevention, and surveillance alerts are configured; the prompt explicitly forbids manipulative conduct.
- [] **MNPI controls:** Regulation FD / selective-disclosure handling, restricted-list integration, expert-network logging rules where applicable, and alternative-data diligence are live.
- [] **Testing:** code changes require independent testing, segregated test environments, documented test results, and a defined disable/rollback process.
- [] **Backtesting:** all results are net of commissions, fees, borrow/funding, slippage, market impact, and realistic latency assumptions; walk-forward and leakage controls are applied.
- [] **Validation:** overfitting tests, scenario analysis, regime segmentation, stress tests, and shadow-mode paper trading are completed.
- [] **Venue readiness:** exchange/broker certification and conformance testing are complete where required; production order-entry paths are tested end-to-end.
- [] **Auditability:** every decision stores the policy version, model version, data snapshot IDs, compliance state, and final action.
- [] **Monitoring:** live dashboards cover PnL, risk, limit usage, data lag, reject rate, slippage, message rate, and unexplained behavior; alerts route to humans.
- [] **Continuous learning:** retraining occurs only in shadow/champion-challenger mode, with documented promotion criteria and rollback triggers.
- [] **Third-party resiliency:** cloud, model-provider, and data-provider dependencies have failover, redundancy, and incident procedures.

Risk control layer	HFT market making	Trend following	News-driven trading
Pre-trade price/size collars	Mandatory hard gate	Mandatory hard gate	Mandatory hard gate
Message-rate / throttle controls	Critical hard gate	Usually low importance	Moderate if event bursts occur
Inventory / position cap	Critical hard gate	Not primary	Moderate
Gross and net exposure limits	Mandatory	Mandatory	Mandatory
Sector/country/ concentration limits	Low to moderate	Critical	Critical
VaR / expected shortfall	Useful but secondary to inventory and intraday limits	Core portfolio gate	Core portfolio gate
Stop-loss / time stop	Recommended internal control	Recommended internal control	Strongly recommended

Risk control layer	HFT market making	Trend following	News-driven trading
Self-trade prevention / wash-trade checks	Critical	Mandatory	Mandatory
Restricted-list / MNPI screen	Mandatory	Mandatory	Critical
Circuit breaker / venue health checks	Critical	Mandatory	Mandatory
Kill switch / flatten procedure	Critical	Critical	Critical
Human escalation	On severe anomalies only	On threshold breaches	On ambiguous or unverifiable events

This matrix is a synthesis of official requirements and sound practice rather than a claim that every item is a universal legal mandate in every jurisdiction. The hard regulatory center of gravity is: pre-trade limits, prevention of erroneous orders, controls over DEA, monitoring, records, testing, and anti-manipulation rules; expected shortfall is included here because it is the tail-risk measure favored in the modern market-risk framework. ²³

Validation tests and metrics

A production trading agent should be treated like a **model-risk program**, not just a backtest. Three of the most important academic warnings are that data snooping can create false discovery, the “best” backtest may simply be overfit, and ordinary Sharpe ratios can be materially inflated by multiple testing and non-normality. Those warnings are directly relevant to prompt selection, because an agent prompt can itself become part of the search space. The validation plan therefore needs statistical controls, realistic execution assumptions, operational tests, and post-deployment monitoring. ²⁴

Validation test	What it checks	Recommended metric or pass condition
Historical backtest with fees and impact	Basic economic viability	Positive edge net commissions, fees, funding, borrow, slippage, and market impact
Purged walk-forward / embargoed CV	Leakage control in time-series validation	Stable out-of-sample performance across folds; no label leakage
Probability of Backtest Overfitting	Whether selected strategy is likely overfit	PBO sufficiently low relative to promotion threshold
Deflated Sharpe Ratio	Whether Sharpe survives multiple testing and non-normality	DSR comfortably above internal significance threshold

Validation test	What it checks	Recommended metric or pass condition
White Reality Check	Whether “best model” really beats benchmark after data snooping	Reject null only after snooping adjustment
Latency-and-slippage replay	Execution realism	Implementation shortfall, fill ratio, reject rate, and queue-loss within tolerance
Regime segmentation	Robustness across macro/volatility states	No catastrophic failure in high-vol / crisis / broken-correlation regimes
Stress and scenario tests	Tail and operational resilience	Within drawdown, ES, and kill-switch tolerances under predefined scenarios
Shadow / paper trading	Live path correctness without capital risk	Stable market-data alignment, routing, logging, and signal behavior
Exchange/broker conformance	Transport and venue correctness	Required certification complete; no protocol failures
Chaos / outage drills	Fail-safe behavior	Correct halt, cancel, flatten, and alert behavior under feed/broker/cloud failure
Drift monitoring	Post-launch degradation	Trigger thresholds for feature drift, hit-rate decay, drawdown acceleration, or rising rejects

This test stack is consistent with the literature on overfitting and data snooping, with FINRA’s emphasis on independent testing, segregated test environments, and post-implementation monitoring, with ESMA/RTS-6 style conformance and stress expectations, and with CME’s certification tooling and algorithm-certification procedures. ²⁵

Metric family	Common production metrics	Variant-specific additions
Return quality	Net PnL, annualized return, Sharpe, Sortino, Calmar, hit rate	Trend: exposure-adjusted return, signal decay; News: event surprise calibration
Tail and portfolio risk	Max drawdown, 1-day VaR, expected shortfall, stress loss, concentration usage	HFT: inventory variance, worst intraday inventory excursion
Execution quality	Implementation shortfall, spread paid/earned, slippage, reject rate	HFT: fill ratio, adverse selection, queue loss, cancel-to-fill, message-rate headroom
Stability and reliability	Uptime, data lag, alert count, exception count, unexplained actions	News: source precision/recall, entity-linking accuracy, confirmation lag

Metric family	Common production metrics	Variant-specific additions
Compliance and audit	Restricted-list hits prevented, policy exceptions, incomplete logs, missing snapshots	HFT: self-trade prevention events; News: MNPI / source-verification escalations

Common risk and execution metrics are standard practice; implementation shortfall remains one of the most useful real-world execution metrics because it measures the gap between decision price and achieved execution, while expected shortfall is the more appropriate tail metric when the aim is to control losses beyond ordinary VaR. ²⁶

The deployment flow below is the recommended promotion path: research, validated backtest, shadow mode, limited capital, champion/challenger monitoring, and only then scale-up. This is also where prompt changes belong: prompt revisions should be versioned, shadow-tested, and promoted exactly like model or strategy changes. ²⁷

```

flowchart TD
    A[Research hypothesis] --> B[Data quality and provenance checks]
    B --> C[Net-cost historical backtest]
    C --> D[Purged CV / walk-forward]
    D --> E[PBO / DSR / Reality Check]
    E --> F[Stress and regime tests]
    F --> G[Conformance and integration tests]
    G --> H[Shadow / paper trading]
    H --> I[Limited-capital canary]
    I --> J[Champion / challenger comparison]
    J --> K[Scale up]
    K --> L[Continuous monitoring]
    L --> M[Alert / rollback / retrain gate]
    M --> H

```

Example interactions and failure-mode mitigations

The examples below show the intended behavior of the prompt family. The important pattern is that the agent is expected to **decline trades** frequently when data quality, risk headroom, or compliance certainty is inadequate.

Example interaction using the core prompt

```

INPUT
{
  "instrument": "ESM6",
  "signals": {"trend_z": 1.4, "event_score": 0.1},
  "risk_state": {"portfolio_var_after": 0.94, "var_limit": 0.90, "gross_after":
1.82, "gross_limit": 2.00},

```

```

    "data_quality": {"fresh": true, "complete": true, "aligned": true,
"provenance_ok": true},
    "compliance_state": {"restricted_list_clear": true, "mmpi_clear": true},
    "execution": {"spread_bps": 0.6, "impact_bps": 1.7}
}

```

OUTPUT

```

{
  "decision": "NO_TRADE",
  "instrument": "ESM6",
  "side": "NONE",
  "size": 0,
  "order_type": "NONE",
  "limit_price": null,
  "time_in_force": "NONE",
  "thesis": "Trend is positive, but projected portfolio VaR breaches the hard
limit after cost-adjusted sizing.",
  "expected_edge_bps": 2.1,
  "confidence": 0.67,
  "holding_period": "5d",
  "risk": {
    "trade_var": 0.05,
    "portfolio_var_after": 0.94,
    "expected_shortfall_after": 1.08,
    "gross_after": 1.82,
    "net_after": 0.54,
    "drawdown_state": 0.63
  },
  "compliance": {
    "approved_instrument": true,
    "approved_venue": true,
    "restricted_list_clear": true,
    "mmpi_clear": true,
    "market_abuse_clear": true
  },
  "data_quality": {
    "fresh": true,
    "complete": true,
    "aligned": true,
    "provenance_ok": true
  },
  "execution_constraints": {
    "max_participation": 0.10,
    "price_collar_ok": true,
    "size_collar_ok": true
  },
  "alerts": ["risk_limit_headroom_low"],
  "audit": {

```

```

    "policy_version": "2026-04-prod-17",
    "model_versions": ["trend_v9", "risk_v4"],
    "data_snapshot_ids": ["md_88421", "risk_21994"],
    "timestamp_utc": "2026-04-29T19:41:00Z"
  }
}

```

Example interaction using the news-driven variant

INPUT

```

{
  "entity": "XYZ Corp",
  "headline": "XYZ exploring strategic alternatives",
  "source_count": 1,
  "primary_filing_present": false,
  "novelty": 0.42,
  "market_reaction": {"5m_return": 6.8, "spread_widening_bps": 24},
  "mnpi_screen": "uncertain"
}

```

OUTPUT

```

{
  "decision": "ESCALATE",
  "event": {
    "entity": "XYZ Corp",
    "type": "strategic_alternatives_rumor",
    "direction": "MIXED",
    "novelty": 0.42,
    "source_quality": 0.38
  },
  "trade_plan": null,
  "time_stop": null,
  "reason": "Single-source rumor, no primary filing, widened spreads, and unresolved MNPI risk. Waiting is required.",
  "audit": {
    "policy_version": "news_v5",
    "data_snapshot_ids": ["news_30018", "md_88428"],
    "timestamp_utc": "2026-04-29T19:42:00Z"
  }
}

```

Example interaction using the HFT market-making supervisor

INPUT

```

{

```



```

    "inventory": {"current": 820, "cap": 1000},
    "venue_health": {"packet_loss": 0.7, "reject_rate": 0.05, "data_sync": false},
    "microstructure": {"spread_ticks": 2, "toxicity_score": 0.81},
    "message_rate": {"current": 8600, "cap": 9000}
  }

```

OUTPUT

```

{
  "decision": "WITHDRAW",
  "policy_update": {
    "spread_bps": 0.0,
    "bid_size": 0.0,
    "ask_size": 0.0,
    "inventory_target": 0.0,
    "max_message_rate": 7000
  },
  "reason": "Data desynchronization plus elevated reject rate and high toxicity near inventory cap. Withdraw and await clean sync.",
  "inventory_state": "high_long",
  "venue_health": "degraded",
  "audit": {
    "policy_version": "mm_sup_v3",
    "timestamp_utc": "2026-04-29T19:42:30Z"
  }
}

```

Failure mode	Typical symptom	Prompt-level mitigation	System-level mitigation
Stale or corrupted data	Impossible prices, missing fields, bad timestamps, desync	Mandatory freshness and completeness checks; force <code>NO_TRADE</code>	Clock sync, feed arbitration, gap detection, failover
Hallucinated or weak textual inference	Confident but wrong event interpretation	Require source allowlist, entity verification, novelty threshold, reason tied to observable facts	Retrieval grounding, citation to primary source, human escalation on ambiguity
MNPI / licensing contamination	Unclear provenance, red flags in alt-data workflow	Hard <code>mnpi_clear</code> and <code>provenance_ok</code> gates	Legal review, vendor diligence, recurring re-onboarding, restricted lists
Backtest overfitting	Great in-sample results, poor OOS stability	Prompt versioning and shadow tests before promotion	PBO, DSR, Reality Check, purged CV, champion/challenger

Failure mode	Typical symptom	Prompt-level mitigation	System-level mitigation
Regime change	Trend signal decay, adverse news reversals, correlation spikes	Require regime/status checks and lower confidence in unfamiliar states	Regime segmentation, drift monitoring, automatic de-risking
Latency blow-up	Reject spikes, queue loss, venue desync	Force <code>WITHDRAW</code> , <code>HALT</code> , or <code>NO_TRADE</code> when latency/health thresholds fail	Co-location/direct-connect, message throttles, heartbeat checks
Third-party concentration / outage	Cloud/model/data-provider disruption	Do not continue on missing critical services	Multi-provider redundancy, disaster runbooks, manual fallback
Audit gap	Decision cannot be reconstructed	<code>audit</code> object mandatory in every output	Immutable logs, snapshot IDs, WORM retention where required

These mitigations are directly supported by the sources most relevant to production operation: NIST’s governance playbook on monitoring, auditing, incident response, transparency, and third-party risk; IOSCO’s warnings on data quality, explainability, concentration, outsourcing, and model/data risks; SEC guidance on alternative-data procedures and MNPI controls; FINRA’s mandated emphasis on testing, disablement, and post-implementation monitoring; and the academic literature on data snooping and overfitting. ²⁸

Open questions and limitations

Several critical items remain intentionally parameterized because the user left them open-ended: profit target, formal risk appetite, leverage, holding period, portfolio construction method, broker/venue stack, approved asset universe, and legal/compliance scope for crypto spot venues. Those values should be supplied by the portfolio manager, risk function, and counsel before the prompt is put into production.

The report’s regulatory synthesis is strongest for U.S./EU/UK market-access and algorithmic-trading controls, plus global IOSCO-style principles. It is weaker on asset-class-specific edge cases in spot crypto, OTC FX, and venue-specific local rules, where firms should maintain allowlists and jurisdiction-specific rulebooks rather than relying on a universal prompt alone. IOSCO’s 2025 AI report also makes clear that concentration, outsourcing, and third-party dependencies remain unresolved supervisory concerns in AI-heavy capital-markets use cases. ²⁹

Finally, the banking-agency model-risk update from the Office of the Comptroller of the Currency ³⁰, the Federal Reserve ³¹, and the FDIC is highly relevant as a governance template, but it explicitly states that generative AI and agentic AI are novel and rapidly evolving and are **not** within the scope of that 2026 guidance. That does not weaken the core recommendation here; it strengthens the case for conservative design, shadow promotion, and explicit ownership until more specific supervisory standards for agentic

trading systems mature. Tail-risk measurement in this report therefore continues to lean on broader market-risk practice, including expected shortfall as reflected in the framework published by the Bank for International Settlements ³² . ³³

¹ ⁶ ⁷ ⁸ ⁹ ¹² ¹³ ¹⁹ ²⁰ ²² ²³ ³⁰ <https://www.law.cornell.edu/cfr/text/17/240.15c3-5>

<https://www.law.cornell.edu/cfr/text/17/240.15c3-5>

² ⁵ ¹¹ ²⁷ ²⁸ <https://airc.nist.gov/airmf-resources/playbook/govern/>

<https://airc.nist.gov/airmf-resources/playbook/govern/>

³ ²⁵ <https://www.finra.org/rules-guidance/notices/15-09>

<https://www.finra.org/rules-guidance/notices/15-09>

⁴ ¹⁰ <https://arxiv.org/abs/2408.06361>

<https://arxiv.org/abs/2408.06361>

¹⁴ ³¹ <https://www.sec.gov/files/code-ethics-risk-alert.pdf>

<https://www.sec.gov/files/code-ethics-risk-alert.pdf>

¹⁵ ²⁹ <https://www.iosco.org/library/pubdocs/pdf/IOSCOPD788.pdf>

<https://www.iosco.org/library/pubdocs/pdf/IOSCOPD788.pdf>

¹⁶ ²¹ <https://arxiv.org/abs/1804.04216>

<https://arxiv.org/abs/1804.04216>

¹⁷ <https://www.sec.gov/submit-filings/about-edgar>

<https://www.sec.gov/submit-filings/about-edgar>

¹⁸ <https://www.cmegroup.com/market-data.html>

<https://www.cmegroup.com/market-data.html>

²⁴ <https://www.ssc.wisc.edu/~bhansen/718/White2000.pdf>

<https://www.ssc.wisc.edu/~bhansen/718/White2000.pdf>

²⁶ ³² <https://www.cis.upenn.edu/~mkearns/finread/impshort.pdf>

<https://www.cis.upenn.edu/~mkearns/finread/impshort.pdf>

³³ <https://www.occ.treas.gov/news-issuances/bulletins/2026/bulletin-2026-13.html>

<https://www.occ.treas.gov/news-issuances/bulletins/2026/bulletin-2026-13.html>